

## 2023 Calculus II Supplement Homework (1)

1. Suppose a sequence  $\{a_n\}$  is bounded and  $\lim_{n \rightarrow \infty} b_n = 0$ . Prove that  $\lim_{n \rightarrow \infty} a_n b_n = 0$ .

2. Find the following limits.

(a)  $\lim_{n \rightarrow \infty} \frac{1}{n} \sin n \cos n^2$ .      (b)  $\lim_{n \rightarrow \infty} (a^n + b^n)^{\frac{1}{n}}$  ( $a > 0, b > 0$ ).

(c)  $\lim_{n \rightarrow \infty} \frac{n^c}{a^n}$  ( $a > 1, c > 0$ ).      (d)  $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{2} + \cdots + \frac{1}{n}\right)^{\frac{1}{n}}$ .

(e)  $\lim_{n \rightarrow \infty} \left(\frac{n}{n+c}\right)^n$  (any  $c$ )      (f)  $\lim_{n \rightarrow \infty} \left(\sqrt{n + \sqrt{n + 2\sqrt{n}}} - \sqrt{n}\right)$ .

3. Determine if the following sequences converge or not and give your reasons. If so, find their limit.

(a)  $a_n = \sin n$ .      (b)  $a_n = n^{(-1)^n}$ .      (c)  $a_n = \cos(\sqrt{4n^2 + 1} \pi)$ .

4. Let  $a_1 \in (0, \frac{\pi}{2})$  and  $a_{n+1} = \sin a_n$ . Does the sequence  $\{a_n\}$  converge? Why? If so, find its limit.

5. Textbook Sec.10.1, Exercise 133 in page 586.